

WEEK 1 — PYTHON BASICS CHEAT SHEET

Light Sensor | Data Science with ESP32 | Grade 8 | Keep this page — refer to it anytime!

1 VARIABLES

```
# Store a number (float)
lux = 48.5

# Store text (string)
level = "Bright"

# Store whole number (int)
raw = 2048

# Change value anytime
lux = 133.0
```

RULES: No spaces | Use underscore: raw_value

2 DATA TYPES

int	2048	Whole number
float	48.57	Decimal number
str	"Bright"	Text in quotes
bool	True	True or False

`type(lux)` returns <class float>

3 PRINT STATEMENTS

```
# Simple
print(lux)

# With label
print("Lux:", lux)

# f-string (BEST!)
print(f"Lux: {lux} lx")

# Multiple values
print(f"{lux} | {level}")
```

USE f-strings: f"text {variable} more text"

4 IMPORTS

```
# At TOP of every file:
import serial
import serial.tools.list_ports
import matplotlib.pyplot as plt
from datetime import datetime
import csv, os

# Install ONCE in terminal:
pip install pyserial matplotlib
```

RULE: Imports go at the VERY TOP of file

5 COMPARISONS

>	Greater than	lux > 80
<	Less than	lux < 30
>=	Greater or equal	lux >= 80
<=	Less or equal	lux <= 30
==	Exactly equal	val == 0
!=	Not equal	val != 0

6 IF / ELIF / ELSE

```
lux = 48.5

if lux > 80:
    level = "Bright"
elif lux > 30:
    level = "Medium"
else:
    level = "Dark"

# One-line shortcut:
level = "Bright" if lux>80 \
    else "Medium" if lux>30 \
    else "Dark"
```

INDENT 4 spaces after if/else or ERROR!

7 WHILE LOOP (preview)

```
# Keep reading forever:
while True:
    line = ser.readline()
    line = line.decode("utf-8")
    line = line.strip()
    if "Light" in line:
        print(line)
```

Full loops in Week 2!

8 FIND ESP32 PORT

```
PORT = None
for port in list_ports.comports():
    if "CH340" in port.description:
        PORT = port.device

# PORT is now "COM5" automatically!
```

FULL LIGHT SENSOR SCRIPT

```
import serial
import serial.tools.list_ports
from datetime import datetime

BAUD = 9600

PORT = None
for port in list_ports.comports():
    if "CH340" in port.description:
        PORT = port.device

print("Found:", PORT)

ser = serial.Serial(PORT, BAUD)

while True:
    line = ser.readline()
    line = line.decode("utf-8",
        errors="ignore")
    line = line.strip()
    if "Light Intensity" in line:
        try:
            lux = float(
                line.split(":")[1])
            now = datetime.now()
            now = now.strftime(
                "%H:%M:%S")
            if lux > 80:
                level = "Bright"
            elif lux > 30:
                level = "Medium"
            else:
                level = "Dark"
            print(f"{now} | ",
                f"{lux} lx | {level}")
        except:
            pass
```

COMMON ERRORS

IndentationError
Fix: Add 4 spaces after if/while

ModuleNotFoundError
Fix: pip install pyserial matplotlib

SerialException
Fix: Close Arduino IDE first!

ValueError
Fix: Wrap code in try/except

THE 3 GOLDEN RULES FOR EVERY LESSON:

1. Upload matching Arduino code to ESP 32 FIRST
2. Close Arduino IDE completely before running Python
3. Only ONE program can use COM5 at a time — never both!